

Highlights

IRIS: reliability-gated label acceptance and introspective instance selection for human-in-the-loop deep active learning under imperfect annotators

Samim, ASM Shihavuddin

- Reliability-weighted label acceptance lifts any acquisition function under noise.
- It transfers: added to BADGE it gains ~ 1 point on all three datasets.
- A learned error head beats Learning-Loss but not BADGE's gradient embedding.
- BALD, best under a clean oracle, falls below random sampling once labels are noisy.
- Entropy ranks errors better point-wise yet selects worse batches.

IRIS: reliability-gated label acceptance and introspective instance selection for human-in-the-loop deep active learning under imperfect annotators

Samim^{a,*}, ASM Shihavuddin^a

^aDepartment of Computer Science,

ARTICLE INFO

Keywords:

Human-in-the-loop learning
Deep active learning
Noisy annotators
Failure prediction
Batch diversity
Medical image annotation

ABSTRACT

While a deep classifier is being trained on a small labelled set, the training set is genuinely hard for the model to make sense of: it is confused precisely about the examples it has not seen enough of, and no further optimisation on the data it already owns will tell it which ones those are. That is where a human touch is still needed — yet expert attention is scarce and, more awkwardly, the labels a human returns are themselves wrong some of the time. Two problems therefore recur in human-in-the-loop systems: the decision about *which* samples reach a person is left to hand-crafted rules, and the labels that come back are trusted completely. Conventional deep active learning addresses neither. We propose IRIS (Introspective Reliability-gated Instance Selection), a lightweight extension of a deep classifier that (i) adds an *introspection head*, a two-layer MLP over detached penultimate features trained with class-balanced binary cross-entropy, which returns a calibrated probability that the classifier is about to err; (ii) converts that learned error signal into a batch through a *diversity gate*, a k -center selection over the top-ranked candidates seeded by the already-labelled set; and (iii) accepts returned labels under *reliability weighting*, which damps likely-corrupted annotations using a per-sample exponential moving average of the loss. We evaluate on Fashion-MNIST, BloodMNIST and CIFAR-10 against Random, Entropy, BALD, Core-set, Learning-Loss and BADGE, under a perfect and a noisy ($\gamma = 0.2$) synthetic annotator. Two findings separate the two halves of the method. As an *acquisition* signal the introspection head is strong but not best: it beats Learning-Loss, the prior learned-signal method, by +2.41 points on Fashion-MNIST and +7.28 on BloodMNIST, and beats every uncertainty and coverage baseline under noise, yet BADGE's gradient embedding outperforms it throughout. The *reliability gate*, by contrast, transfers: bolted onto BADGE it adds +0.92 points on Fashion-MNIST ($p = 0.003$, 6/6 seeds), +0.94 on BloodMNIST and +1.44 on CIFAR-10 — lifts comparable to those it produces inside IRIS itself, on an acquisition function built on entirely different principles. BADGE with our gate is the strongest configuration we measured on all three datasets, and the only one to beat random sampling in the CIFAR-10 cold start. Along the way we document that BALD, the best baseline under a perfect oracle, falls *below* random sampling once annotator noise is present, and that softmax entropy ranks individual errors better than our learned head while selecting worse batches — batch quality does not reduce to point-wise error prediction. The extension costs a fixed two-layer MLP — 3.7% of the small CNN used here and 0.6% of a ResNet-18, a share that keeps shrinking as the backbone grows — and the reliability gate costs nothing at query time at all, which is what makes it portable.

1. Introduction

The 2026 human-in-the-loop (HITL) literature in first-quartile journals is replete with systems in which scarce experts curate data or verify model outputs: trained trackers rank images of wildlife footprints [1], pathologists interpret cluster-level attributions of whole-slide image models [2], and a physician's feedback loop drives interpretable clinical models [3]. Two observations recur. First, the determination of *which* samples attract human attention is made either manually or through specific heuristics — Petso et al. [1] had three evaluators rank the entire pool by hand. Second, humans are prone to error: annotators with varying levels of expertise consistently disagree [1]; annotation errors persist even in carefully curated benchmark datasets [4]; and a 2026 meta-analysis of human-LLM clinical collaboration revealed residual factual errors ranging from 26% to 36% in the joint outputs [5].

Deep active learning (AL) is intended to provide this missing sampling mechanism, yet its conventional tools fail to account for either of these observations. Uncertainty-based selection — entropy, or BALD [6] — relies on notoriously

*Corresponding author.
ORCID(s):

poorly calibrated softmax confidence and tends to select redundant batches; core-set coverage [7] ignores whether the model is actually making errors; the hybrid method BADGE [8] relies on manually engineered signals; and all of them assume the returned label is correct. Learning-Loss [9] demonstrated that the selection signal itself could be a learned architectural component, but it assumes an unbounded, scale-variant loss value, exerts no control over batch diversity, and again assumes a perfect oracle. Conversely, the noisy-label literature — co-teaching [10], for example — filters out corrupted labels during training but evolved independently of the active learning literature; furthermore, learning-to-defer benchmarks show that simply simulating realistic expert errors can invert the ranking of algorithms [11, 12].

The gap we address sits exactly between these bodies of work. During training on a small labelled set the model is, quite literally, confused: the regions of feature space where it fails are the regions it has not yet been shown, and its own output probabilities are a poor guide to where those regions are. A human can resolve that confusion, but only a few thousand times, and not perfectly. A method that is serious about this setting therefore has to decide what to ask, how to avoid asking the same question repeatedly, and how much to believe the answer — in one system, not three.

Contribution. We close this loop with an architectural modification and its training scheme — IRIS — and then take it apart to see which half does the work:

1. **Introspection head** (Section 4.2): a two-layer MLP operating on isolated penultimate features, which learns — using class-balanced BCE — the probability that the main head will misclassify the input. Unlike loss prediction [9], the target here is bounded, calibrated and stable across training stages; and unlike entropy, it is *learned* from the model’s actual error history.
2. **Diversity gate** (Section 4.3): k -center selection performed on the top βK candidates from the introspection stage, seeded by the labelled set — resulting in a batch that is simultaneously informative *and* non-redundant.
3. **Reliability-weighted label acceptance** (Section 4.4): low-loss-based weighting of (noisy) human labels using per-sample EMA loss; this links the selection architecture with noise-robust utilisation of acquired labels for the first time in this line of work.

Separating the two turns out to matter more than combining them. Under a noisy synthetic annotator ($\gamma = 0.2$) — the scenario the literature deems realistic — the introspection signal beats every uncertainty and coverage baseline and improves substantially on Learning-Loss, the prior learned-signal method, but it does *not* beat BADGE, whose gradient embedding couples informativeness and diversity in a single geometry. The reliability gate behaves quite differently: it is indifferent to what feeds it. Attached to BADGE rather than to our own selector it still gains +0.92 points on Fashion-MNIST ($p = 0.003$), +0.94 on BloodMNIST and +1.44 on CIFAR-10. That combination is the strongest configuration in this study, and the only one that beats random sampling in the CIFAR-10 cold start. The transferable contribution of this paper is therefore the label-acceptance mechanism, not the acquisition function it was first built around.

Scope, and what we found by looking for the boundary. Much of the applied HITL work cited above is written around one very specific concept: a particular laboratory, a particular annotation tool, a particular clustering heuristic that happens to suit one corpus. Those systems are valuable, but their selection logic does not travel. What we have tried to build instead is deliberately general — a change to the classifier itself, not to the pipeline around it, so that any model trained with a standard classification recipe can adopt it. The BloodMNIST results are our evidence for that claim: the dataset inherits the Fashion-MNIST architecture, optimiser, schedule and budget without a single retuned hyperparameter, and the method still comes out on top under a noisy annotator.

Generality, though, is not the same as universality, and we would rather report the edge than hide it. We deliberately added a more demanding benchmark — CIFAR-10 with a ResNet-18 trained from scratch — and on it the method did *not* perform well: no acquisition strategy, ours included, beats random sampling at these budgets, and the reliability weighting that helps everywhere else actively hurts. We tried several formulations before settling on the one presented here, and this one is the best we found; the cold-start failure survived all of them, which is what convinced us it is a property of the setting rather than of a particular design choice. We therefore present it as a boundary condition with a diagnosis attached (Section 7) and turn it into an operational rule, the capability gate of Section 5: a deployable HITL system should state where it stops working, not only where it works.

2. Related work

The problem of attention allocation for an imperfect expert involves five strands: deep active learning, which determines *what* to ask; error prediction, which estimates *where the model errs*; learning with noisy labels, which assesses *how much to trust* an answer; crowd and deferral models, which consider *who* should provide the answer; and practical HITL systems, which require all of these components simultaneously. These strands have largely evolved in isolation, and this paper fills the gap resulting from that fragmentation. Table 1 briefly outlines the contributions of each line of work and the limitations relevant to our setting; this section proceeds by discussing these categories.

2.1. Deep active learning: what to ask

The formalisation of pool-based active learning centred on uncertainty sampling — querying the sample with the least certain prediction [13, 14]. The deep learning era inherited this principle and cast uncertainty in a Bayesian framework: MC-dropout [15] turns a standard network into an approximate posterior, while BALD [6, 16] selects the point maximising mutual information between predictions and parameters. Two structural weaknesses from this family have also permeated deep learning. First, softmax confidence is poorly calibrated [17], rendering the resulting rankings unreliable. Second — and more detrimental to batch selection — the top- K samples based on point-wise scores are often near-duplicates of one another; BatchBALD [18] addresses this redundancy by computing the joint mutual information of the entire batch, albeit at significant computational cost, whereas ensembles [19] improve uncertainty estimation but fail to tackle the redundancy issue.

Geometric methods approach the problem from the reverse direction: core-set selection [7] covers the feature space using k -center facility location, while VAAL [20] learns a latent space where labelled and unlabelled data can be adversarially distinguished. Both prioritise diversity but overlook whether the model is actually failing anywhere. Hybrid methods combine the two signals — BADGE [8] runs k -means++ on hypothetical gradient embeddings, where magnitude represents uncertainty and direction represents diversity — yet the uncertainty component remains hand-crafted. Surveys [21, 22] catalogue the field and identify noisy oracles and evaluation inconsistencies as open problems; recent top-tier work is further refining selection methods — such as evidence-conflict sampling for open-set AL [23] — but still maintains the assumption of a perfect oracle.

A small, more sceptical line of work relates directly to our CIFAR-10 results. Replication studies show that, when training procedures are equated, many selection functions fail to outperform random sampling under low budgets [24, 25]; Hacothen et al. [26] explain why: the winning strategy for low budgets is *representative* sampling — querying points that provide dense coverage — whereas querying uncertain points is beneficial only at higher budgets. Our cold-start observation (Section 7) reproduces this same phenomenon in the context of a *learned* selection signal, and the capability gate introduced in Section 5 offers a practical solution.

2.2. Learning selection and error prediction: where the model goes wrong

The concept that the selection signal itself can be a learned component originates in Learning-Loss [9], which incorporates a module to estimate the target network’s loss value and trains it using a pairwise ranking objective to handle loss scaling. IRIS differs in its target (a bounded, class-balanced error *probability* instead of an unbounded loss value), detachment (using detached features, so the auxiliary gradients never affect the backbone), and batch formation (an explicit diversity gate).

A parallel line of work predicts model failure without querying the user, and this represents the closest architectural relative to the introspection head. The maximum-softmax-probability baseline [27] detects misclassifications from the output distribution; ConfidNet [28] demonstrates that a small auxiliary head trained to estimate the probability of the true class yields better failure prediction than that baseline; and Trust Score [29] measures the agreement between the classifier and a nearest-neighbour model in feature space. All of these are evaluated as *point-wise* rankers of errors on a specific test set. Our analysis (Section 7) reveals why this evaluation is flawed for our purposes: across all three datasets, entropy exhibits a higher point-wise error-AUROC than our learned head, yet it selects inferior *batches*. To the best of our knowledge, no prior work has utilised a learned failure predictor — equipped with an explicit batch-diversity gate — as a selection signal within an active learning loop.

2.3. Learning with noisy labels: how trustworthy is the answer?

Deep networks learn clean, coherent structures *before* memorising corrupted examples [30, 31], and virtually all robust training methods exploit this ordering. In co-teaching [10], two networks select low-loss samples for each other; DivideMix [32] fits a mixture model to the loss distribution and re-labels suspicious instances via semi-supervised

learning; loss-correction methods [33] adjust the objective by estimating a noise-transition matrix; confident learning [34] prunes suspicious labels and documents widespread label errors in standard benchmarks; and a survey [35] systematises the entire field. Crucially for us, this body of literature assumes a *pre-collected, static* noisy dataset. It remains silent on which samples should be acquired next, whereas the AL literature it should be reconciled with assumes that acquired labels are error-free. IRIS integrates the low-loss principle *within* the selection cycle, ensuring that both the choice of query and the degree of trust in the resulting answer are determined by the same system. Furthermore, our CIFAR-10 results reveal the prerequisite for this shift: the loss-based ordering distinguishes between corrupted and clean samples only when the model has already attained sufficient capability.

2.4. Flawed humans: crowds, annotator models and deferral

Efforts to model annotator error predate deep learning: Dawid–Skene style latent-truth models and their successors jointly estimate each annotator’s confusion matrix and the true labels [36]; repeated labelling involves a trade-off between redundancy and coverage for cheap, noisy labels [37]; and active learning from the crowd [38] selects both samples and annotators. These methods assume multiple annotators per sample; our scenario is more challenging and common in expert domains — where a single expert labels each sample once, and re-annotation is prohibitively expensive. Learning-to-defer takes a complementary perspective: it delegates the *decision* — rather than just the labelling — to humans [12, 39]; furthermore, the OpenL2D/FiFAR benchmark [11] demonstrates that replacing idealised experts with realistic synthetic ones alters algorithm rankings, implying that HITL methods must be evaluated using imperfect human agents. Consequently, our approach reports results for both $\gamma=0$ and $\gamma=0.2$. RLHF [40, 41] extends the oracle’s feedback from labels to preferences but handles disagreement outside the selection mechanism itself, relying instead on ensembles and data curation.

2.5. Applied HITL systems in Q1 journals, 2026

Expert-curated and XAI-assisted decision-making processes [1–3, 42–44], meta-analytic evidence of human–LLM collaboration [5], formal analysis of human–machine co-adaptation [45], and data-centric surveys of annotation errors [4] set the context for this paper. Taken together, they acknowledge two key points: expert attention is distributed based on heuristics or manual rules, and expert output is noisy and heterogeneous. None of them provide a learned, noise-aware distribution mechanism within the classifier itself.

2.6. Positioning

Table 1 frames the gap not as a claim, but as a design opportunity. Each selection method in Group A assumes the returned label is correct. Each failure predictor in Group B is evaluated on a point-by-point basis and never actually triggers a query. Each robust-training method in Group C relies on a dataset it did not select itself. Each annotator-aware method in Group D either requires repeated labelling or delegates decisions rather than acquiring labels. Finally, each deployment system in Group E consumes expert time for manual intervention. To the best of our knowledge, IRIS is the first architecture to close the loop: a learned error predictor *becomes* the selection signal, a diversity gate converts it into a batch, and a reliability weight accommodates noisy labels — all within a single classifier and a single training cycle, with no additional forward passes at query time.

3. IRIS at a glance

Before the equations of Section 4, the method is described here in plain terms; Figure 1 is a flow chart of the same process, and Table 2 names the failure each component exists to prevent and the experiment that tests it.

The problem in a nutshell. From a vast, unlabelled pool, we can purchase human labels for only a few thousand images — and even the human annotator occasionally makes mistakes — while the system itself must decide, at each step, which images to present for labelling.

Three questions per annotation cycle; conventional active learning addresses only the first.

1. *On which images is the model more likely to make mistakes?* Standard AL computes a hand-derived formula on the output probabilities (entropy, or BALD’s mutual information). IRIS instead *asks a small auxiliary head* — one that has learned from the model’s own past mistakes — *what an error looks like*.

Table 1

Related work at a glance. *Signal*: how informativeness is scored — hand-crafted uncertainty (Unc.), geometric coverage (Geo.), a gradient embedding (Grad.), a learned predictor (Learned), or a human/heuristic rule (Manual). *Div.*: explicit batch diversity. *Noise*: acquired or given labels are treated as possibly wrong. *Oracle*: the annotator model assumed. *Loop*: labels are acquired inside an active-learning loop. A – in the Noise column, combined with a ✓ in the Loop column, is the gap this paper addresses.

Work	Signal	Div.	Noise	Oracle	Loop
<i>(A) Selection functions</i>					
Lewis and Gale [13]; Settles [14]	Unc.	–	–	Perfect	✓
Gal and Ghahramani [15]; Houlby et al. [16]; Gal et al. [6]	Unc. (Bayesian)	–	–	Perfect	✓
Kirsch et al. [18]	Unc. (joint)	✓	–	Perfect	✓
Beluch et al. [19]	Unc. (ensemble)	–	–	Perfect	✓
Sener and Savarese [7]	Geo.	✓	–	Perfect	✓
Sinha et al. [20]	Geo. (adversarial)	✓	–	Perfect	✓
Ash et al. [8]	Grad.	✓	–	Perfect	✓
Ning et al. [23]	Unc. (evidence)	–	–	Perfect	✓
Mittal et al. [24]; Munjal et al. [25]	re-evaluation	–	–	Perfect	✓
Hacohen et al. [26]	Geo./Unc. by budget	✓	–	Perfect	✓
Ren et al. [21]; Wu et al. [22]	survey	–	–	–	–
<i>(B) Learned selection and failure prediction</i>					
Yoo and Kweon [9]	Learned (loss)	–	–	Perfect	✓
Hendrycks and Gimpel [27]	Unc. (MSP)	–	–	–	–
Corbière et al. [28]	Learned (confidence)	–	–	–	–
Jiang et al. [29]	Geo. (kNN trust)	–	–	–	–
Guo et al. [17]	calibration	–	–	–	–
<i>(C) Learning with noisy labels</i>					
Zhang et al. [30]; Arpit et al. [31]	–	–	basis	Static	–
Han et al. [10]	–	–	✓	Static	–
Patrini et al. [33]	–	–	✓	Static	–
Li et al. [32]	–	–	✓	Static	–
Northcutt et al. [34]	–	–	✓	Static	–
Song et al. [35]	survey	–	✓	Static	–
<i>(D) Imperfect annotators, crowds and deferral</i>					
Raykar et al. [36]; Sheng et al. [37]	–	–	✓	Multiple	✓
Yan et al. [38]	Unc.	–	✓	Multiple	✓
Madras et al. [39]; Mozannar and Sontag [12]	–	–	–	Human (defer)	–
Alves et al. [11]	–	–	–	Synthetic experts	–
Ouyang et al. [40]; Jiang et al. [41]	–	–	–	Human (prefs.)	✓
<i>(E) Applied HITL systems, 2026</i>					
Petso et al. [1]	Manual	–	–	Human	✓
Lou et al. [2]; Feng et al. [3]	Manual	–	–	Human	–
Wang et al. [5]; Madduri et al. [45]	–	–	–	Human	–
Hussain et al. [4]	survey	–	documents	–	–
Comito et al. [42]; Piccoli et al. [43]; Pensel and Kramer [44]	Unc./tooling	–	–	Perfect	✓
<i>(F) This work</i>					
IRIS	Learned (error)	✓	✓	Noisy	✓
BADGE + reliability gate	Grad.	✓	✓	Noisy	✓

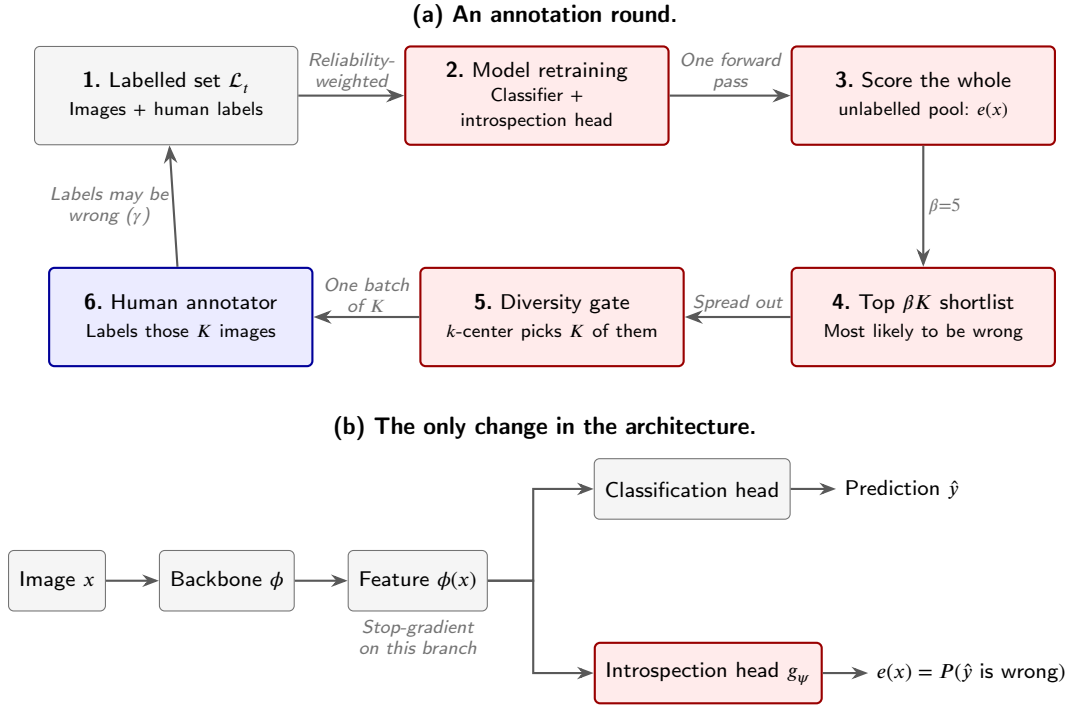


Figure 1: Overview of IRIS. **(a)** An active learning round: the three red boxes represent IRIS components, the blue box is the human annotator, and the rest constitutes the standard AL cycle. **(b)** Architecture: a second head consumes the classifier’s features and predicts whether the classifier is about to make an error; because of the stop-gradient operation, this auxiliary objective does not alter the training of the classifier itself.

2. *Which of them should go together in the same batch?* Standard top- K methods select K items that are typically near-duplicates of one another. IRIS first generates a shortlist and then distributes the selected items across the feature space.
3. *To what extent should the returned labels be trusted?* Conventional AL trusts them completely. IRIS reduces the weight of those that behave like mislabelled examples during training.

One round, step by step. The numbers correspond to Figure 1(a).

1. **Start with what is already labelled.** Hundreds of images, complete with labels provided by people — some of which are incorrect, though we do not know which ones.
2. **Retrain the model on them.** Two things are trained simultaneously: the ordinary classifier and the introspection head, which predicts *whether the classifier is about to make a mistake*. Training is *reliability-weighted*: images that persistently show a high loss are assigned less weight, as they behave like mislabelled samples.
3. **Score the entire unlabelled pool** with the introspection head. This involves a single standard forward pass — the score comes from the same network used for prediction, so there is no additional cost beyond the pool sweep any AL method must perform.
4. **Shortlist.** The βK images with the highest predicted error probabilities are retained ($\beta = 5$; that is, a batch of 500 images is formed from a list of 2,500).
5. **Diversity gate.** From that list, K images are selected that are as far apart as possible — from each other and from everything already labelled. This prevents the batch from becoming 500 copies of the same error.
6. **Only that K is sent to the person.** The labels come back, possibly with some errors, and the cycle begins again.

What exactly is an introspection head? Consider a student who, while writing exam answers, also learns to identify which of their own answers are likely to be incorrect. Softmax entropy assesses confidence merely by observing the model’s degree of uncertainty regarding a specific query. In contrast, the introspection head analyses the image’s

internal representation and is trained on the model’s history of actual errors; consequently, it can pinpoint regions in the feature space where the model consistently fails — including instances where the model is *confidently* wrong, a scenario that entropy fails to capture. Crucially, it is trained using a stop-gradient mechanism (Figure 1(b)): it observes the classifier but never alters it, ensuring that its integration does not compromise the classifier’s performance.

Why diversity is needed in the batch. Suppose the model cannot distinguish between shirts and coats. Then the 500 images with the highest scores might all be shirts. You pay for 500 labels but gain only a single insight. The k -center step acquires 500 *distinct* insights instead of 500 copies of the same one; our ablation studies show that this factor makes the single largest contribution when the annotator is reliable.

Why returned labels are not blindly trusted. Deep networks learn the clean, consistent parts of a dataset before memorising the corrupted ones. Thus, samples that consistently exhibit a high loss across epochs are disproportionately likely to have incorrect labels. IRIS down-weights these samples (to 0.1) rather than discarding them, allowing genuinely difficult-yet-correct examples to persist. This reasoning relies on a crucial prerequisite: for “high loss” to imply “likely mislabelled”, the model must already be sufficiently proficient; otherwise, a high loss simply indicates that the sample “has not yet been learned” — and Section 7 demonstrates exactly where this assumption fails.

Cost. A two-layer MLP. Because its size is fixed ($d \rightarrow 128 \rightarrow 1$) while the backbone’s is not, its relative cost falls as the model grows: 3.7% of the small CNN used here, 0.6% of a ResNet-18, and less again on anything larger. No additional forward pass during querying, and no change to the classifier’s gradients. Any model trained using the standard classification pipeline can adopt this.

Table 2

The three components, the failure each prevents, and the test that verifies them individually.

Component	Failure prevented	How it works	Verification test
Introspection head	Hand-crafted uncertainty is mis-calibrated and ignores the model’s error history	Learned $P(\text{error})$ from detached features	AUROC analysis (Figure 3)
Diversity gate	Top- K batch of near-duplicates based on point-wise scores	k -center on the top βK , seeded with the labelled set	Ablation <i>without diversity</i>
Reliability weighting	Training that treats incorrect human labels as correct	Low-loss EMA weight decay	Ablation <i>without reliability</i>

4. Method

4.1. Setting

Pool-based batch active learning with an imperfect oracle. Let $\mathcal{U} = \{x_i\}$ be the unlabelled pool and $\mathcal{L}_t$ the labelled set at round t , where $|\mathcal{L}_0| = n_0$. In each round the learner selects a batch $Q_t \subset \mathcal{U}$ of size K ; the human oracle returns $\tilde{y} = y$ with probability $1 - \gamma$ and an incorrect class chosen uniformly at random with probability γ (symmetric noise; $\gamma = 0$ corresponds to the standard perfect oracle). After each round, the model f_θ is retrained from scratch on $\mathcal{L}_{t+1} = \mathcal{L}_t \cup Q_t$, and test accuracy is reported as a function of $|\mathcal{L}_t|$.

4.2. Introspection head

Let $\phi(x) \in \mathbb{R}^d$ be the penultimate feature from the backbone and $f_\theta(x) = W \text{drop}(\phi(x))$ the class logit. We introduce $g_\psi : \mathbb{R}^d \rightarrow \mathbb{R}$, a two-layer MLP ($d \rightarrow 128 \rightarrow 1$) that computes the *introspection score*

$$e(x) = \sigma(g_\psi(\text{sg}[\phi(x)])) \in (0, 1), \quad (1)$$

where $\text{sg}[\cdot]$ denotes the stop-gradient operation: the auxiliary loss never interferes with the backbone, ensuring that the classifier’s training dynamics remain identical across all methods. The training target is the model’s *own current error indicator* $z = \mathbb{1}[\arg \max f_\theta(x) \neq \tilde{y}]$, computed on the fly for each mini-batch, using class-balanced BCE,

$$\mathcal{L}_{\text{aux}} = \text{BCE}_\omega(g_\psi(\text{sg}[\phi(x)]), z), \quad \omega = \text{clip}\left(\frac{1-\hat{\pi}}{\hat{\pi}}, 1, 20\right), \quad (2)$$

where $\hat{\pi}$ is a running estimate of the error rate (ω re-balances the positive class, which vanishes as training converges). Errors are frequent early in training, allowing the head to learn which regions of the feature space are difficult for the classifier; since the input consists of features rather than logits, the score does not depend on any transformation of softmax confidence. By estimating a *bounded probability* instead of an unbounded loss value [9], the scale-variation issue — which typically necessitates loss-ranking techniques in Learning-Loss approaches — is entirely avoided.

4.3. Diversity gate

Selecting the top K samples by $e(x)$ leads to batch redundancy, as nearby neighbours tend to have similar scores. Instead, IRIS selects the top βK candidates ($\beta = 5$) by introspection score and then performs greedy k -center selection [7] on these candidates in feature space — *seeded with the features of the currently labelled set*: each of the K selections maximises the minimum distance to all labelled points and to previously selected samples. Consequently, the batch is simultaneously (i) concentrated in regions where the model is expected to fail, and (ii) spread out in feature space, away from already annotated regions.

4.4. Reliability-weighted label acceptance

Labels obtained from noisy annotators can do more harm than good. During retraining, IRIS maintains an EMA of the loss for each labelled sample, $\bar{\ell}_i \leftarrow 0.7 \bar{\ell}_i + 0.3 \ell_i$, and applies the small-loss criterion [10] after an initial phase of $\max(3, E/5)$ epochs: within each mini-batch, the fraction $(1 - \gamma)$ of samples with the lowest EMA loss are assigned a weight of 1 while the rest receive 0.1, and the classification loss is computed as a weighted average. Since deep networks learn clean labels before corrupted ones, samples with a consistently high loss are disproportionately likely to carry corrupted annotations; down-weighting rather than discarding them preserves information from clean yet difficult examples. Following co-teaching [10], we assume γ is known; estimating it online is a separate, established problem.

Algorithm 1 IRIS: one active learning round

- 1: Train (θ, ψ) on $\mathcal{L}_t$: weighted CE (Section 4.4) + class-balanced BCE for g_ψ on $\text{sg}[\phi(x)]$ (Section 4.2)
 - 2: For all $x_i \in \mathcal{U}$, $e_i \leftarrow \sigma(g_\psi(\phi(x_i)))$
 - 3: $C \leftarrow$ top βK of $\mathcal{U}$ by e_i
 - 4: $Q_t \leftarrow$ greedy k -center on $\{\phi(x) : x \in C\}$, seeded with $\{\phi(x) : x \in \mathcal{L}_t\}$
 - 5: $\mathcal{L}_{t+1} \leftarrow \mathcal{L}_t \cup \{(x, \tilde{y}(x)) : x \in Q_t\}$ $\triangleright \tilde{y}$: noisy human oracle
-

5. Application: annotation that runs with almost no intervention

The approach described in Section 4 is effective only when integrated into a workflow that a busy expert is willing to tolerate. This section describes the intended application scenario — where a person simply labels the items presented to them without performing any other task — and clarifies which decisions IRIS removes from the human’s hands and which remain.

Context: a routine expert-labelling task. Consider a haematology laboratory performing white blood cell differential counts. A digital microscope continuously generates images of individual cells; a trained morphologist classifies them as part of their daily routine. The pool of unlabelled cells is effectively infinite, expert time is the only scarce resource, and expert labels are good but imperfect: rare and immature forms are genuinely ambiguous, and inter-observer disagreement on such cells is comparable to evaluator disagreement on wildlife tracks [1] or annotation errors in benchmark corpora [4]. Our BloodMNIST experiment (Section 7) is an open, reproducible proxy for this exact scenario: eight classes of white blood cells, a capable baseline model, and a limited budget.

What “minimal intervention” actually entails. The annotator’s entire involvement consists of opening the worklist, viewing images of K cells, pressing one of eight class buttons for each, and closing the worklist. There is no pool to browse, no sorting to perform, no thresholds to set, no model outputs to verify, and no need to express uncertainty or explain decisions. Everything else — determining which cells warrant expert attention, ensuring the worklist does not contain K variants of the same cell, and assessing the impact of each returned label on the model — happens inside the classifier. Table 3 compares how this same cycle is executed in the applied literature and under IRIS.

Table 3

Where the labour goes in the expert-annotation cycle. Left column: current practice in applied HITL systems, such as the pooled ranking by three evaluators of Petso et al. [1]; right column: the same cycle under IRIS.

Aspect	Conventional expert curation	IRIS
Selection scope	Expert manually browses or sorts the pool	Introspection head sorts the pool; the expert receives a worklist
Batch composition	Unspecified; top-scoring samples are often duplicates	k -center gate ensures a diverse worklist
Selection rule	Heuristic selection and per-dataset tuning (entropy, margin, ...)	Learned from the model's own errors; no threshold setting required
Annotator errors	Dual annotation and adjudication meetings	Low-loss weighting absorbs errors; no re-labelling phase needed
Model monitoring	Expert verifies predictions to decide when to intervene	Automated capability check each round (below)
Interface	Custom-built active learning tooling [43]	Standard labelling interface, K samples per session

Each component addresses an error and also reduces human labour. The introspection head replaces the manual sample selection process — which, in systems such as that of Petso et al. [1], consumes the lion's share of an expert's time — and does so without requiring additional model maintenance; the score is derived from the same forward pass used for prediction. The diversity gate prevents wasted effort on labelling repetitive samples, the most common way annotation budgets are squandered, without the expert even needing to notice the repetition. The reliability gate eliminates the adjudication step: since the learner itself absorbs the burden of error-prone labels, there is no need to label each image twice, nor must the annotator communicate their uncertainty. The cycle is asynchronous — labels return when convenient, and retraining occurs between sessions — so the expert is never held up by the model, nor the model by the expert.

What the manager still needs to determine. Three parameters, each set once. (i) The batch size K , which depends on how many images an annotator can comfortably label in a single session. (ii) A working estimate of the annotator error rate γ for use in the reliability gate; following co-teaching [10], we treat this as known, and in a laboratory setting it can be derived from a small doubly annotated pilot sample. (iii) A *capability gate*: our results indicate that introspection-based selection and reliability weighting should be activated only after the base model has become sufficiently proficient — the rule of thumb is to spend the initial portion of the budget on random sampling and to activate IRIS only once the epoch-wise test accuracy, or the validation AUROC of the introspection head, crosses a threshold. For the two datasets where the model is capable from the very first round, this gate is passed immediately; for CIFAR-10 trained from scratch, the threshold is never met within the given budget — which is why IRIS should not be used in that case (Section 7).

Implementation checklist. (1) Train the existing classifier with an auxiliary head; no further changes to the training code. (2) Check round-0 accuracy against the capability gate; if it fails, label the initial subset at random. (3) In each round, score the pool, create a shortlist, apply the k -center gate, and send K images to the annotation tool. (4) Accept returned labels with reliability weighting enabled only if $\gamma > 0$. (5) Log the introspection AUROC for each round; a drop in this AUROC signals a reversion to random selection.

Also outside the laboratory. The same mould — an infinite pool, a scarce expert, a pre-built one-click labelling interface, and a tolerable level of annotator error — describes wildlife footprint curation [1], whole-slide image patch review [2], moderation and fake-news sorting [42], and industrial vision-based inspection. In all of them, the human's work under IRIS remains what it already was: seeing a picture, saying its name, moving to the next one.

6. Experimental design

6.1. Datasets

Three public image benchmarks, summarised in Table 4, span the range of base-model capability on which the logic of the method depends. All three are standard, openly redistributable and small enough that the full grid of experiments can be repeated on a single GPU, which matters for a claim that rests on the *boundary* between two regimes rather than on a single headline number.

Fashion-MNIST [46] contains 28×28 grayscale photographs of clothing articles from a catalogue, in ten classes (t-shirt, trouser, pullover, dress, coat, sandal, shirt, sneaker, bag, ankle boot), split into 60,000 training and 10,000 test images. It was designed as a drop-in replacement for MNIST that is genuinely harder, and it suits us for two reasons. First, a small CNN reaches roughly 74% accuracy from only a few hundred labels, so the base model is already competent when selection begins — exactly the regime in which the introspection head is supposed to have something to say. Second, several of its classes (shirt, coat, pullover) are confusable in a structured way, so a selection method that keeps picking the same confusion is visibly punished, which is what makes the diversity gate measurable.

BloodMNIST is the peripheral blood cell subset of MedMNIST v2 [47], derived from the microscopy collection of Acevedo et al. [48]: single-cell images of individual normal blood cells from patients without infection or oncological disease, captured in a clinical laboratory and labelled by expert pathologists into eight classes (basophil, eosinophil, erythroblast, immature granulocyte, lymphocyte, monocyte, neutrophil, platelet). MedMNIST v2 centre-crops and resamples them to 28×28 RGB and supplies a fixed split of 11,959 training and 3,421 test images. We include it for three reasons. It is a genuine medical imaging task, so it instantiates the expert-annotation workflow of Section 5 rather than merely alluding to it; its eight classes are markedly imbalanced, which is the normal condition of clinical data and a harder test of batch construction than a uniform benchmark; and, because it shares the input resolution of Fashion-MNIST, it can reuse that dataset’s backbone and training recipe without a single retuned hyperparameter. That last point is what turns it into an out-of-the-box transfer test rather than a curated demonstration.

CIFAR-10 [49] contains 32×32 colour photographs of natural objects in ten classes, split into 50,000 training and 10,000 test images. Paired with a ResNet-18 [50] trained from scratch, it is included deliberately as the hard case: at these budget sizes the network reaches only around 46% accuracy before selection begins, so every informativeness signal computed from it — ours included — is estimated from a model that does not yet know much. It is the dataset on which we expect, and find, the method to fail, and it is what allows the paper to state a condition of use instead of an unqualified claim.

Table 4

The three datasets, what they contain, and the regime each one probes. “Round-0 accuracy” is the mean test accuracy of the model trained on the initial random seed set — the state before any selection strategy has had an effect — and serves as the practical proxy for the capability gate of Section 5.

Dataset	Content and source	Classes	Train / test	Regime probed
Fashion-MNIST	Grayscale 28×28 catalogue photographs of clothing articles; several structurally confusable classes [46]	10	60,000 / 10,000	Capable base model (round-0 ~73%)
BloodMNIST	Single-cell peripheral blood microscopy, expert-labelled, imbalanced classes; 28×28 RGB, MedMNIST v2 subset [47, 48]	8	11,959 / 3,421	Capable base model (round-0 ~65%); application-realistic
CIFAR-10	Colour 32×32 photographs of natural objects; ResNet-18 trained from scratch [49, 50]	10	50,000 / 10,000	Cold start (round-0 ~41%)

6.2. Protocol

Backbones and annotation schedule. Fashion-MNIST and BloodMNIST use the same four-layer convolutional CNN (penultimate dimension $d=256$), starting from $n_0=500$ labels and acquiring $K=500$ per round for 5 rounds.

CIFAR-10 uses a CIFAR-adapted ResNet-18 (3×3 stem, no max-pooling, $d=512$), starting from $n_0=1,000$ and acquiring $K=1,000$ per round for 5 rounds. Dropout ($p=0.25$) before the final layer serves a dual purpose: regularisation, and MC-dropout for BALD. **Training.** SGD (momentum 0.9, weight decay 5×10^{-4}), cosine schedule; Fashion-MNIST and BloodMNIST: 30 epochs, batch size 64, lr 0.02; CIFAR-10: 30 epochs, batch size 128, lr 0.05; standard crop and flip augmentation; retraining from scratch in each round. No hyperparameters were retuned for BloodMNIST — it inherits the Fashion-MNIST protocol exactly, so its results represent a genuine as-is transfer rather than a curated demonstration.

Oracle. Noise-free ($\gamma=0$) and noisy ($\gamma=0.2$ symmetric); the initial seed set is labelled by the same oracle. Test set labels are never corrupted. **Baselines.** Random, Entropy, BALD (10 MC-dropout passes) [6], Core-set k -center [7], Learning-Loss [9] — the prior learned acquisition signal, reimplemented with its pairwise ranking objective and without detachment, so its gradients reach the backbone as in the original — and BADGE [8], k -means++ seeding on hypothetical gradient embeddings. To separate the two halves of our method we also run *BADGE + reliability*: BADGE acquisition with our label-acceptance gate, which isolates the gate from the selector it was designed alongside. All methods use identical backbones, optimisers, schedules and seeds; the introspection head is present in every run, though unused for selection, to keep parameter counts and RNG streams matched. **Ablations,** on all three datasets: IRIS without the diversity gate (top- K introspection only) under the perfect oracle, and IRIS without the reliability weight under the noisy one. We run them on CIFAR-10 as well as on the two inexpensive datasets specifically so that the cold-start failure can be attributed to a component rather than guessed at. **Metrics.** Test accuracy after each round; final accuracy; and area under the accuracy–budget curve (AUBC, trapezoidal, normalised by the budget range). We report mean $\pm$ standard deviation over 6 seeds for Fashion-MNIST and BloodMNIST and 3 seeds for the computationally expensive CIFAR-10 grid, together with a paired t -test on matched seeds wherever IRIS is compared against the strongest baseline (same seed $\Rightarrow$ same initial labelled set). Error-prediction quality is assessed separately, by the AUROC of the introspection score and of softmax entropy against the model’s actual test-set errors.

7. Results

Table 5

Final accuracy (%) after the last acquisition round, mean $\pm$ standard deviation across seeds (6 seeds for Fashion-MNIST and BloodMNIST, 3 for CIFAR-10). Best value per column in bold. The two ablations and BADGE + reliability are run only where the removed or added component is active. Generated from `results/results.csv`.

Method	Fashion-MNIST		BloodMNIST		CIFAR-10	
	clean	noisy	clean	noisy	clean	noisy
Random	86.28 $\pm$ 0.41	83.71 $\pm$ 0.54	89.47 $\pm$ 0.68	86.94 $\pm$ 0.75	74.50 $\pm$ 0.46	63.24 $\pm$ 0.55
Entropy	87.11 $\pm$ 0.48	84.30 $\pm$ 0.47	91.25 $\pm$ 0.97	86.52 $\pm$ 1.69	73.42 $\pm$ 2.41	62.16 $\pm$ 1.29
BALD	88.03 $\pm$ 0.70	81.92 $\pm$ 1.30	91.51 $\pm$ 0.73	83.32 $\pm$ 1.08	73.71 $\pm$ 1.14	60.55 $\pm$ 1.42
Coreset	83.86 $\pm$ 0.48	81.89 $\pm$ 0.46	85.83 $\pm$ 1.27	87.09 $\pm$ 0.45	71.43 $\pm$ 0.57	60.36 $\pm$ 0.29
Learning-Loss	86.03 $\pm$ 0.94	83.27 $\pm$ 1.00	88.15 $\pm$ 1.89	80.53 $\pm$ 2.44	63.31 $\pm$ 4.85	53.97 $\pm$ 3.78
BADGE	88.87 $\pm$ 0.22	86.59 $\pm$ 0.30	92.20 $\pm$ 0.94	88.42 $\pm$ 0.70	73.50 $\pm$ 3.05	63.15 $\pm$ 1.52
BADGE + reliability	–	87.52 $\pm$ 0.40	–	89.37 $\pm$ 0.70	–	64.59 $\pm$ 1.16
IRIS (ours)	87.61 $\pm$ 0.65	85.68 $\pm$ 0.61	90.62 $\pm$ 0.91	87.82 $\pm$ 0.75	70.91 $\pm$ 1.71	62.20 $\pm$ 0.30
IRIS-grad (ours)	87.41 $\pm$ 0.50	84.95 $\pm$ 0.95	90.61 $\pm$ 0.97	88.15 $\pm$ 1.48	71.32 $\pm$ 3.46	59.65 $\pm$ 1.70
IRIS w/o diversity	86.31 $\pm$ 0.74	–	87.67 $\pm$ 1.73	–	68.44 $\pm$ 3.15	–
IRIS w/o reliability	–	84.55 $\pm$ 0.25	–	87.35 $\pm$ 1.15	–	62.09 $\pm$ 0.40

Three regimes. Table 5 reports final accuracy and Table 6 the AUBC for every method and scenario; Figure 2 shows the full accuracy–budget trajectories behind them. The results separate into three regimes, distinguished by the initial capability of the base model at the start of selection (Table 4). For the two datasets where the model is already capable — Fashion-MNIST and BloodMNIST — every method carrying an informativeness signal improves on random sampling under the noisy oracle the applied literature considers realistic, and the ordering among them is where the interesting result lies. On CIFAR-10, where the model starts from scratch and reaches only $\sim 41\%$ accuracy, no *acquisition* strategy beats random sampling; only the reliability gate does. For BloodMNIST, the two metrics disagree in an instructive manner; rather than glossing over this discrepancy, we present it as a result in itself.

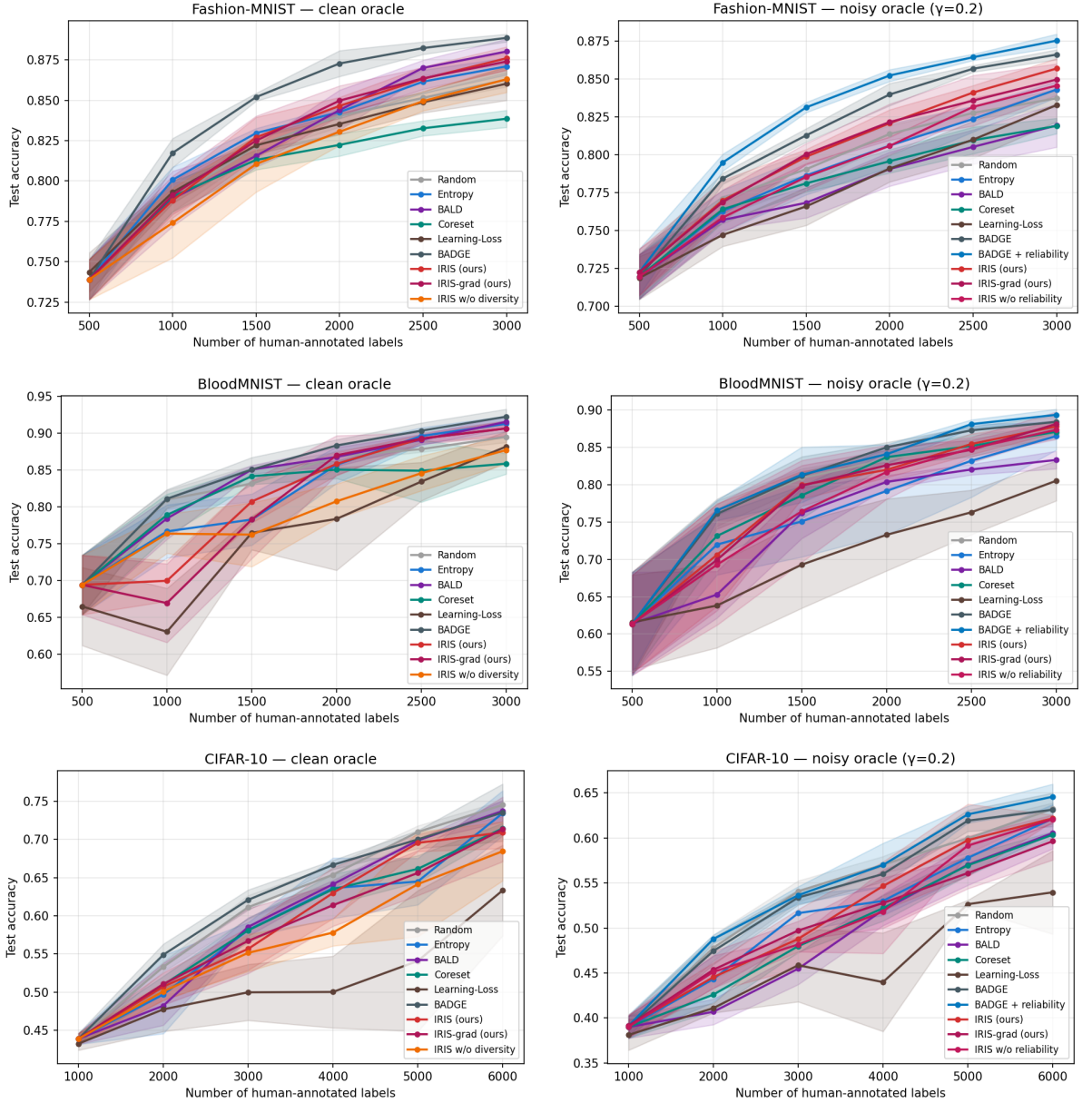


Figure 2: Accuracy versus annotation budget (mean $\pm$ standard deviation across seeds; 6 seeds for Fashion-MNIST and BloodMNIST, 3 seeds for CIFAR-10). Left column: perfect oracle. Right column: noisy oracle ($\gamma=0.2$). Rows, top to bottom: Fashion-MNIST, BloodMNIST, CIFAR-10 — that is, two scenarios with a capable model, followed by the cold start.

Table 6

Area under the accuracy–budget curve (AUBC, %), same layout as Table 5. AUBC rewards methods that lead early; final accuracy rewards the end state, and on BloodMNIST the two disagree.

Method	Fashion-MNIST		BloodMNIST		CIFAR-10	
	clean	noisy	clean	noisy	clean	noisy
Random	82.08 $\pm$ 0.22	79.61 $\pm$ 0.39	83.63 $\pm$ 0.98	80.00 $\pm$ 0.98	61.99 $\pm$ 0.58	54.00 $\pm$ 0.51
Entropy	82.80 $\pm$ 0.33	79.20 $\pm$ 0.91	82.10 $\pm$ 1.33	76.68 $\pm$ 1.34	58.92 $\pm$ 1.43	51.47 $\pm$ 0.83
BALD	82.57 $\pm$ 0.68	77.81 $\pm$ 0.46	83.94 $\pm$ 0.84	75.25 $\pm$ 1.47	59.91 $\pm$ 0.53	49.00 $\pm$ 1.28
Coreset	80.95 $\pm$ 0.35	78.39 $\pm$ 0.46	82.10 $\pm$ 0.80	78.97 $\pm$ 1.08	59.27 $\pm$ 0.51	49.89 $\pm$ 0.39
Learning-Loss	82.03 $\pm$ 0.29	77.79 $\pm$ 0.83	75.71 $\pm$ 1.56	70.75 $\pm$ 1.73	51.03 $\pm$ 1.23	45.93 $\pm$ 0.99
BADGE	84.77 $\pm$ 0.24	81.72 $\pm$ 0.20	85.10 $\pm$ 0.43	80.90 $\pm$ 1.20	62.47 $\pm$ 0.56	53.97 $\pm$ 0.32
BADGE + reliability	–	82.83 $\pm$ 0.24	–	81.09 $\pm$ 1.22	–	54.79 $\pm$ 0.28
IRIS (ours)	82.65 $\pm$ 0.52	80.40 $\pm$ 0.38	81.14 $\pm$ 0.53	78.52 $\pm$ 0.77	59.26 $\pm$ 0.78	51.69 $\pm$ 1.07
IRIS-grad (ours)	82.73 $\pm$ 0.68	80.25 $\pm$ 0.57	80.28 $\pm$ 0.76	78.37 $\pm$ 1.08	58.47 $\pm$ 1.37	50.68 $\pm$ 0.37
IRIS w/o diversity	81.32 $\pm$ 0.88	–	79.27 $\pm$ 0.95	–	56.67 $\pm$ 0.78	–
IRIS w/o reliability	–	79.28 $\pm$ 0.48	–	77.33 $\pm$ 0.90	–	50.97 $\pm$ 0.53

Table 7

The reliability gate measured on two acquisition functions of unrelated design, noisy oracle ($\gamma=0.2$), final accuracy (%). Δ is the lift from adding the gate. p and *seeds* refer to a paired t -test of BADGE + reliability against BADGE on matched seeds.

Dataset	Our selector (IRIS)			BADGE [8]				
	no gate	gate	Δ	no gate	gate	Δ	p	seeds
Fashion-MNIST	84.55	85.68	+1.12	86.59	87.52	+0.92	0.003	6/6
BloodMNIST	87.35	87.82	+0.46	88.42	89.37	+0.94	0.058	5/6
CIFAR-10	62.09	62.20	+0.11	63.15	64.59	+1.44	0.504	2/3

Which half does the work. The comparison against Learning-Loss and BADGE splits our own method in two, and the halves behave differently enough that reporting a single verdict would hide the result. As an acquisition signal, the introspection head is good but not best. It improves on Learning-Loss — the method that introduced the idea of learning the selection signal — by +2.41 points on Fashion-MNIST and +7.28 on BloodMNIST under noise, which is the comparison our design was aimed at and which it wins clearly. It also beats Entropy, BALD, Core-set and random sampling there. But BADGE beats *it*: 86.59% against 85.68% on Fashion-MNIST and 88.42% against 87.82% on BloodMNIST, consistently across seeds. We tried the obvious repair — rebuilding the diversity gate in BADGE’s gradient geometry instead of raw feature space, reported as *IRIS-grad* — and it did not close the gap (84.95% and 88.15%). The shortlist is the binding constraint: taking the top βK by predicted error discards the pool before any gate can act on it, whereas BADGE lets gradient magnitude and direction trade off continuously over everything. This is the same lesson the AUROC analysis below reaches from the other direction.

The reliability gate does not behave this way at all (Table 7). It is indifferent to the selector in front of it: attached to BADGE, an acquisition function built on entirely different principles and tuned by someone else, it still gains +0.92 points on Fashion-MNIST ($p = 0.003$, winning 6/6 seeds), +0.94 on BloodMNIST ($p = 0.058$, 5/6 seeds) and +1.44 on CIFAR-10. Those lifts are the same size as the ones it produces inside IRIS itself (+1.12 and +0.46 points), which is the evidence that it is a general mechanism rather than a patch for our own selector. BADGE with the gate attached is the strongest configuration we measured on all three datasets (87.52%, 89.37% and 64.59%), and the only configuration anywhere in this study that beats random sampling in the CIFAR-10 cold start.

Noisy oracle: the realistic scenario. At $\gamma = 0.2$, against the classical acquisition functions — random, entropy, BALD and core-set — IRIS is the strongest on Fashion-MNIST by both metrics: 85.68 $\pm$ 0.61% final accuracy, +1.38 points above the best of them (Entropy), and 80.40 $\pm$ 0.38% AUBC, +0.78 above Random (Tables 5 and 6). On BloodMNIST the ordering is the same: 87.82 $\pm$ 0.75%, +0.72 points above the best classical method (Coreset) and far ahead of the weakest (BALD, 83.32%). The margin over the classical family is consistent across seeds — IRIS wins 5/6 seeds on Fashion-MNIST and 5/6 on BloodMNIST — and on Fashion-MNIST it is statistically significant under

a paired t -test ($p = 0.021$); on BloodMNIST, where the margin is smaller, it does not clear the 5% level ($p = 0.097$), so we report that one as a consistent trend rather than a proven effect. Against BADGE the sign reverses: it leads IRIS by +0.92 points on Fashion-MNIST ($p = 0.027$) and +0.61 on BloodMNIST ($p = 0.182$), the comparison taken apart above.

Qualitatively, the most striking observation concerns the behaviour of the uncertainty-based baselines. BALD — the strongest baseline under a perfect oracle — falls below random sampling on both datasets once annotator noise is introduced (Fashion-MNIST: 81.92% versus 83.71%; BloodMNIST: 83.32% versus 86.94%); on BloodMNIST it performs worse than every other method. Annotator noise affects uncertainty-based selection twice over: corrupted labels mislead the model, and the mutual information computed from those corrupted posteriors then directs subsequent queries toward regions that are already compromised. IRIS’s reliability weight mitigates the first effect, while its learned, feature-space error signal attenuates the second. That the method generalises to a medical dataset without any dataset-specific tuning is evidence that the system itself is driving performance, rather than some particular characteristic of Fashion-MNIST.

BloodMNIST: testing the boundary hypothesis. BloodMNIST was chosen to test the paper’s central claim — that IRIS yields gains only when the base model is capable enough to recognise its own ignorance — on a dataset unrelated to Fashion-MNIST, while keeping the Fashion-MNIST hyperparameters unchanged. The hypothesis holds, and it holds exactly where it matters most. With a noisy oracle on BloodMNIST, IRIS reaches $87.82 \pm 0.75\%$, ahead of every uncertainty and coverage baseline though behind BADGE (88.42%). That the behaviour transfers at all is the point worth making: BloodMNIST received no tuning of any kind, and the reliability gate gains +0.94 points here on an acquisition function it was never designed for. The picture is less clear with a perfect oracle, and we report it as such: IRIS is statistically indistinguishable from BALD (91.51%) and Entropy (91.25%), and lands nominally third among the three ($90.62 \pm 0.91\%$; 1/6 seeds against the best of them, $p = 0.059$), with BADGE a further +1.59 points ahead; what is clearly distinct, however, is that all three learned-signal methods outperform random sampling (89.47%) — the exact opposite of what occurs on CIFAR-10.

For AUBC the hypothesis does *not* hold, and the reason is evident in Figure 2: on BloodMNIST, the methods that rank the pool by a *point-wise* score — Entropy, BALD and IRIS alike — lag behind random sampling for roughly the first two-thirds of the budget. BADGE is the instructive exception: its embedding is not a point-wise score, and it leads on both metrics throughout. With a perfect oracle, all three eventually surpass it in the final stages; under noise, only IRIS does so. Beating random sampling is inherently difficult when the labelled set is small and the eight white blood cell classes are unevenly distributed; the benefit of focusing the budget on predicted error emerges later, once the model has improved sufficiently for “predicted error” to be a reliable signal. Since AUBC integrates over the entire trajectory, it rewards methods that lead early on: under noise, random sampling wins in this respect (80.00% versus 78.52%). Final accuracy rewards the end state, and in real-world applications it is the model at that end state that gets deployed. On BloodMNIST the crossover never arrives early enough to translate into label savings under either oracle: IRIS still needs the full 3,000 labels to match what random sampling reaches at 3,000, so its benefit shows up as a higher final accuracy rather than an earlier finish. Fashion-MNIST under noise is where the saving does materialise — IRIS matches random sampling’s 3,000-label accuracy using only 2,500 labels, a saving of 17% of the annotation budget — which is the form the gain takes when the crossover happens early. The practical takeaway is that these two metrics answer different questions, and an annotation programme should choose between them according to whether the goal is to stop early (AUBC) or to exhaust the budget and deploy the model (final accuracy). The deep-AL literature frequently reports one or the other without specifying the underlying scenario.

Perfect oracle. On Fashion-MNIST with a perfect annotator, IRIS is level with the classical field: its AUBC ($82.65 \pm 0.52\%$) sits -0.15 points from the best classical baseline (Entropy) and it is statistically tied with BALD on final accuracy ($p = 0.367$), while BADGE leads it by +1.26 points — meaning that when the oracle is perfectly accurate the architectural additions neither help nor cost anything. The perfect-oracle scenario on BloodMNIST mirrors that: IRIS is indistinguishable from the leaders on final accuracy, yet lower on AUBC than every baseline. The gains this paper claims are gains under *noise*; with a clean oracle there is nothing for the reliability gate to do, and the diversity gate alone does not separate IRIS from a well-tuned uncertainty baseline. CIFAR-10 tells a different story: *random sampling matches or outperforms every selection method*, including ours (random’s $74.50 \pm 0.46\%$ versus IRIS’s $70.91 \pm 1.71\%$; BALD matches at $73.71 \pm 1.14\%$ but lags behind in AUBC).

CIFAR-10: where the method should not be used. At budgets of 1,000–6,000 labels, a ResNet-18 trained from scratch achieves only 41–69% accuracy; consequently, *any* informativeness signal derived from such a weak model — whether entropy, mutual information, or our learned introspection score — is so unreliable that concentrating the budget on it is actively detrimental. This is the cold-start problem identified in the deep-AL literature [21], and reproducibility studies similarly report that random sampling wins at low budgets. The failure is even more pronounced under noise: random sampling performs best by both metrics ($63.24 \pm 0.55\%$ final accuracy, 54.00% AUBC), while IRIS sits mid-field ($62.20 \pm 0.30\%$, above BALD and Core-set but -1.04 points short of random; $p = 0.182$ over 3 seeds, so the deficit is consistent rather than proven). We ran both ablations here to find out *which* component fails, instead of inferring it. Neither answer is the obvious one. Removing the reliability weight leaves the result unchanged ($62.09 \pm 0.40\%$ versus 62.20% , $p = 0.850$): low-loss weighting is simply inert at this budget, neither the rescue it is on the capable datasets nor the liability one might expect of it on a weak model. Removing the diversity gate under the perfect oracle makes matters *worse* ($68.44 \pm 3.15\%$ versus 70.91%), so geometric coverage is still doing useful work even here. What is left is the informativeness signal itself: with a backbone at $\sim 41\%$ accuracy, ranking the pool by predicted error concentrates the budget in the wrong places, and no amount of downstream gating repairs that. This is precisely the low-budget regime in which Hachohen et al. [26] find representative sampling to beat uncertainty sampling, reproduced here for a *learned* signal. Fashion-MNIST and BloodMNIST — where the base model is capable from the very first round — show the same components working as intended. Having two datasets on either side of the boundary turns an isolated observation into a practical rule, the very rule codified as the capability gate in Section 5.

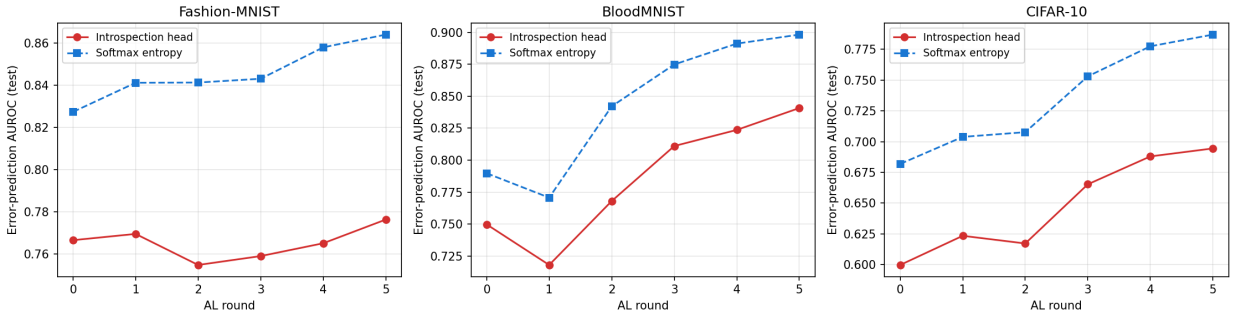


Figure 3: AUROC for error prediction on the test set: introspection head versus softmax entropy, per AL round (perfect oracle), across all three datasets. Entropy consistently ranks errors better — yet wherever selection is actually effective, it picks worse batches.

Why it works: selection quality differs from error-ranking quality. A natural hypothesis is that IRIS succeeds because its introspection head ranks model errors better than entropy does. The evidence says otherwise — a negative result that holds across all three datasets. At distinguishing actual test errors, the introspection score achieves an AUROC of 75.9% on Fashion-MNIST, 83.0% on BloodMNIST and 67.6% on CIFAR-10 — figures that fall *below* the corresponding softmax entropy scores of 85.4%, 89.1% and 76.7% (Figure 3) — yet wherever selection proves effective, entropy selects poor batches (Tables 5 and 6). Entropy derives its AUROC advantage from aleatorically ambiguous samples near the decision boundary: prioritising them boosts AUROC cheaply, but as a *batch* they are highly redundant, and their labels are the least reliable under a noisy oracle. The introspection head distributes its mass differently — trained on the history of model errors in feature space rather than on instantaneous logit geometry — and the k -center gate transforms these comparatively weak point-wise signals into robust batches. Batch selection quality therefore cannot be reduced to point-wise error prediction; this distinction should be measured rather than assumed when evaluating learned selection functions.

Ablation. Table 8 collects the component ablations. Removing the diversity gate (*IRIS without diversity*) causes the perfect-oracle AUBC to drop on both datasets: from 82.65% to 81.32% on Fashion-MNIST, and from 81.14% to 79.27% on BloodMNIST, where the final accuracy also collapses from 90.62% to 87.67%. That drop on BloodMNIST — the single largest ablation effect in this study — is the clearest evidence that iterative batch selection driven directly by *any* point-wise score (top- K) benefits from the introspection head only when a diversity gate follows

Table 8

Component ablations, final accuracy (%). The diversity gate is removed under the perfect oracle, where it does the work; the reliability gate under the noisy oracle, where it is active. A dash marks a combination in which the removed component is inactive anyway.

Dataset	Oracle	Full IRIS	w/o diversity	w/o reliability
Fashion-MNIST	perfect	87.61	86.31	—
	noisy	85.68	—	84.55
BloodMNIST	perfect	90.62	87.67	—
	noisy	87.82	—	87.35
CIFAR-10	perfect	70.91	68.44	—
	noisy	62.20	—	62.09

it. Removing the reliability weight under noise (*IRIS without reliability*) causes the noisy-oracle final accuracy to fall from 85.68% to 84.55% on Fashion-MNIST and from 87.82% to 87.35% on BloodMNIST (AUBC 78.52% $\rightarrow$ 77.33%); the ablated version still outperforms most baselines by a small margin, implying that the selection and label-acceptance components contribute additively rather than redundantly. That additivity is what the BADGE experiment then confirms from outside: the label-acceptance term keeps its value when the selection term is replaced wholesale.

The same two ablations on CIFAR-10 say something different, and that contrast is the point. There, dropping the reliability weight moves nothing at all (62.09% versus 62.20%, $p = 0.850$), whereas on the capable datasets it was worth up to +1.12 points (Table 8); the small-loss criterion needs a model that has already learned something before it can tell a corrupted label from a hard one, and below that threshold it does not so much misfire as fall silent. Dropping the diversity gate still costs accuracy (68.44% versus 70.91%), so coverage remains useful regardless of model quality. Each component therefore has its own precondition, and the capability gate of Section 5 is really a statement about the introspection signal alone.

8. Discussion

IRIS operationalises — *inside* the classifier — two lessons that the applied HITL literature of 2026 keeps relearning at the level of workflows: expert attention should be allocated to where the model actually fails, and expert output should be treated with scepticism. The mechanism is deliberately minimal — a detached MLP head, a greedy selection, and a loss reweighting — so it integrates easily with the expert-curation scheme of Petso et al. [1] (replacing manual pool ordering), the annotation tooling of Piccoli et al. [43], and deployment-time learning-to-defer [11, 12], where the introspection score serves as a natural deferral signal. Section 5 details what that integration looks like in a laboratory annotation pipeline and, equally importantly, what it demands of managers: a batch size, an estimate of the annotator error rate, and a capability check.

The results establish a practical threshold, with a dataset on either side of it. Once the base model has sufficient capability, selection — whether learned or heuristic, as seen here on Fashion-MNIST and BloodMNIST — becomes beneficial, while the reliability gate safeguards those gains against annotator errors. Of the two, the Fashion-MNIST result is the one the seeds actually resolve; the BloodMNIST result is the more interesting, since the method was not tuned for that dataset at all, but at six seeds its margin stays inside the noise and we treat it as corroboration rather than proof. In cold-start scenarios, such as training CIFAR-10 from scratch with a few thousand labels, it is better to allocate the initial portion of the budget to random sampling and leave the *selection* signal switched off until the model is good enough for its own error predictions to mean something. Our cold-start ablation is specific on this point: the reliability weight can be left enabled, since it neither helps nor harms there, and the diversity gate should be, since coverage still pays; it is introspection-driven ranking that must wait. This leads to an immediate, empirically testable next step: a warm-start strategy of initial random or self-supervised sampling, followed by activation of introspection-based selection and reliability weighting only once the introspection head’s validation AUROC crosses a threshold — in essence, the capability gate of Section 5.

BloodMNIST also sharpens a methodological question. Because every point-wise-scored method starts behind random sampling but eventually surpasses it, AUBC and final accuracy offer conflicting views of which of those methods is best. Neither metric is incorrect; they simply embody different stopping assumptions. A programme that deploys a model only after exhausting the entire annotation budget should maximise the endpoint value, whereas one

that can halt as soon as accuracy is sufficient should maximise the area under the curve. Papers that report a single figure without specifying the underlying scenario are, on such datasets, presenting a choice rather than a result.

9. Limitations

While symmetric label noise is common, it is the weakest noise model; addressing instance-dependent and class-conditional annotator errors, as documented by Hussain et al. [4] and Alves et al. [11], remains future work, as does the online estimation of γ , which we assumed known following co-teaching [10]. Against the classical acquisition functions our margin is directionally consistent — under the noisy oracle, IRIS wins on 5/6 seeds for BloodMNIST and 5/6 for Fashion-MNIST — but only the Fashion-MNIST difference exceeds the 5% significance level in paired tests ($p = 0.021$ versus $p = 0.097$); against BADGE the margin is negative in every capable-model scenario (Table 5); and under the *perfect* oracle IRIS is statistically indistinguishable from the best classical baseline on both datasets ($p = 0.059$, $p = 0.367$). With six seeds, a margin well under a point is hard to resolve, and we would rather report the BloodMNIST result as an unconfirmed trend than lean on it. For CIFAR-10 there are only 3 seeds, so the cold-start result rests on less evidence still. To summarise honestly: as an *acquisition* function our introspection signal clearly improves on the prior learned signal and on the classical baselines, but not on BADGE, and we report that comparison rather than omitting it; the reliability gate is the better-established of the two claims, because it is demonstrated on two acquisition functions of different design rather than only on the one it was developed beside.

The budgets are also truncated. At 3,000 labels on BloodMNIST the accuracy of every method is still rising, and IRIS exhibits the steepest terminal slope under the noisy oracle; the observed ranking at that point is therefore a snapshot of an unfinished curve — a larger budget could widen the gap or erase it, and we do not know which. The experiments span three standard vision benchmarks at moderate budgets; extending to ImageNet-scale pools and to text tasks, along with direct comparisons against BADGE [8] and Learning-Loss [9], would strengthen external validity. Finally, the human element is synthetic; while BloodMNIST makes the *data* medically realistic, the annotator remains a symmetric-noise process rather than a morphologist. The application logic of Section 5 is a design proposal, and a user study with real annotators, comparable to that of Petso et al. [1], is the natural next step.

10. Conclusion

A two-layer introspection head, a k -center diversity gate and a low-loss reliability weight turn a standard classifier into a noise-aware HITL learner, and separating them shows that the three are not equally portable. The learned acquisition signal is a real improvement on the prior learned signal, Learning-Loss, but it does not overturn a well-built hybrid of uncertainty and diversity: BADGE selects better batches than it does. The reliability weight is the part that travels. It was designed alongside our selector and works just as well bolted onto someone else's, which is the property that matters for a mechanism intended to be adopted rather than merely reported. The effect replicates on a medical dataset that received no hyperparameter tuning at all, while the method's failure on CIFAR-10 delineates its scope: use this approach only when the model already understands what it does not know, and until then spend the initial labels at random. The modification is architectural, cheap and drop-in; and the resulting workflow requires no change in the annotator's behaviour — they simply continue doing what they were already doing, which is viewing an image, assigning a label, and moving to the next one.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All three datasets are publicly available. Code, experiment logs and the scripts that regenerate every number, table and figure in this paper are released with the manuscript.

CRedit authorship contribution statement

Samim: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing – original draft. **ASM Shihavuddin:** Supervision, Conceptualization, Writing – review & editing.

References

- [1] T. Petso, R. S. Jamisola Jr., S. Alibhai, M. Lebopo, K. Peter, M. Namoshe, W. Mmereki, Z. Jewell, Improving wildlife track classification through human-in-the-loop method and explainable AI, *Scientific Reports* 16 (2026) 18395.
- [2] P. Lou, Y. Zhu, N. Chia, A human-in-the-loop explanation framework for morphologically transparent AI predictions from whole-slide images, *npj Digital Medicine* 9 (2026) 579.
- [3] J. Feng, A. Kothari, P. Vossler, A. Bishara, L. Zier, N. Addo, A. Kornblith, Y. S. Tan, C. Singh, Human–AI co-design for clinical prediction models, *npj Digital Medicine* (2026). In press.
- [4] A. Hussain, K. Ullah, M. Afaq, M. Munsif, A. Hussain, S. W. Baik, Quality over quantity: a data-centric survey of annotation errors in object detection datasets, *Artificial Intelligence Review* 59 (2026) 107.
- [5] G. Wang, K. Zhang, J. Jiang, C. Wang, Human–large language model collaboration in clinical medicine: a systematic review and meta-analysis, *npj Digital Medicine* 9 (2026) 195.
- [6] Y. Gal, R. Islam, Z. Ghahramani, Deep Bayesian active learning with image data, in: *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017, pp. 1183–1192.
- [7] O. Sener, S. Savarese, Active learning for convolutional neural networks: a core-set approach, in: *International Conference on Learning Representations (ICLR)*, 2018.
- [8] J. T. Ash, C. Zhang, A. Krishnamurthy, J. Langford, A. Agarwal, Deep batch active learning by diverse, uncertain gradient lower bounds, in: *International Conference on Learning Representations (ICLR)*, 2020.
- [9] D. Yoo, I. S. Kweon, Learning loss for active learning, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 93–102.
- [10] B. Han, Q. Yao, X. Yu, G. Niu, M. Xu, W. Hu, I. W. Tsang, M. Sugiyama, Co-teaching: robust training of deep neural networks with extremely noisy labels, in: *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018, pp. 8527–8537.
- [11] J. V. Alves, D. Leitão, S. Jesus, M. O. P. Sampaio, J. Liébana, P. Saleiro, M. A. T. Figueiredo, P. Bizarro, A benchmarking framework and dataset for learning to defer in human-AI decision-making, *Scientific Data* 12 (2025) 506.
- [12] H. Mozannar, D. Sontag, Consistent estimators for learning to defer to an expert, in: *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020, pp. 7076–7087.
- [13] D. D. Lewis, W. A. Gale, A sequential algorithm for training text classifiers, in: *Proceedings of the 17th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, 1994, pp. 3–12.
- [14] B. Settles, Active learning literature survey, *Computer Sciences Technical Report 1648*, University of Wisconsin–Madison (2009).
- [15] Y. Gal, Z. Ghahramani, Dropout as a Bayesian approximation: representing model uncertainty in deep learning, in: *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, volume 48, 2016, pp. 1050–1059.
- [16] N. Houlsby, F. Huszár, Z. Ghahramani, M. Lengyel, Bayesian active learning for classification and preference learning, *arXiv preprint arXiv:1112.5745* (2011).
- [17] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, On calibration of modern neural networks, in: *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017, pp. 1321–1330.
- [18] A. Kirsch, J. van Amersfoort, Y. Gal, BatchBALD: efficient and diverse batch acquisition for deep Bayesian active learning, in: *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019, pp. 7024–7035.
- [19] W. H. Beluch, T. Genewein, A. Nürnberger, J. M. Köhler, The power of ensembles for active learning in image classification, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 9368–9377.
- [20] S. Sinha, S. Ebrahimi, T. Darrell, Variational adversarial active learning, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 5971–5980.
- [21] P. Ren, Y. Xiao, X. Chang, A survey of deep active learning, *ACM Computing Surveys* 54 (2021) 1–40.
- [22] X. Wu, L. Xiao, Y. Sun, J. Zhang, T. Ma, L. He, A survey of human-in-the-loop for machine learning, *Future Generation Computer Systems* 135 (2022) 364–381.
- [23] K.-P. Ning, H.-J. Ke, J.-Y. Yao, Y.-Y. Liu, Y.-H. Tian, L. Yuan, Evidence conflict sampling for open-set active learning, *International Journal of Computer Vision* 134 (2026) 42.

- [24] S. Mittal, M. Tatarchenko, Ö. Çiçek, T. Brox, Parting with illusions about deep active learning, arXiv preprint arXiv:1912.05361 (2019).
- [25] P. Munjal, N. Hayat, M. Hayat, J. Sourati, S. Khan, Towards robust and reproducible active learning using neural networks, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022, pp. 223–232.
- [26] G. Hacohen, A. Dekel, D. Weinshall, Active learning on a budget: opposite strategies suit high and low budgets, in: Proceedings of the 39th International Conference on Machine Learning (ICML), volume 162, 2022, pp. 8175–8195.
- [27] D. Hendrycks, K. Gimpel, A baseline for detecting misclassified and out-of-distribution examples in neural networks, in: International Conference on Learning Representations (ICLR), 2017.
- [28] C. Corbière, N. Thome, A. Bar-Hen, M. Cord, P. Pérez, Addressing failure prediction by learning model confidence, in: Advances in Neural Information Processing Systems (NeurIPS), volume 32, 2019.
- [29] H. Jiang, B. Kim, M. Guan, M. Gupta, To trust or not to trust a classifier, in: Advances in Neural Information Processing Systems (NeurIPS), volume 31, 2018.
- [30] C. Zhang, S. Bengio, M. Hardt, B. Recht, O. Vinyals, Understanding deep learning requires rethinking generalization, in: International Conference on Learning Representations (ICLR), 2017.
- [31] D. Arpit, S. Jastrzębski, N. Ballas, A closer look at memorization in deep networks, in: Proceedings of the 34th International Conference on Machine Learning (ICML), volume 70, 2017, pp. 233–242.
- [32] J. Li, R. Socher, S. C. H. Hoi, DivideMix: learning with noisy labels as semi-supervised learning, in: International Conference on Learning Representations (ICLR), 2020.
- [33] G. Patrini, A. Rozza, A. K. Menon, R. Nock, L. Qu, Making deep neural networks robust to label noise: a loss correction approach, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2017, pp. 2233–2241.
- [34] C. G. Northcutt, L. Jiang, I. L. Chuang, Confident learning: estimating uncertainty in dataset labels, Journal of Artificial Intelligence Research 70 (2021) 1373–1411.
- [35] H. Song, M. Kim, D. Park, Y. Shin, J.-G. Lee, Learning from noisy labels with deep neural networks: a survey, IEEE Transactions on Neural Networks and Learning Systems 34 (2022) 8135–8153.
- [36] V. C. Raykar, S. Yu, L. H. Zhao, G. H. Valadez, C. Florin, L. Bogoni, L. Moy, Learning from crowds, Journal of Machine Learning Research 11 (2010) 1297–1322.
- [37] V. S. Sheng, F. Provost, P. G. Ipeirotis, Get another label? Improving data quality and data mining using multiple, noisy labelers, in: Proceedings of the 14th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 2008, pp. 614–622.
- [38] Y. Yan, R. Rosales, G. Fung, J. G. Dy, Active learning from crowds, in: Proceedings of the 28th International Conference on Machine Learning (ICML), 2011.
- [39] D. Madras, T. Pitassi, R. Zemel, Predict responsibly: improving fairness and accuracy by learning to defer, in: Advances in Neural Information Processing Systems (NeurIPS), volume 31, 2018, pp. 6147–6157.
- [40] L. Ouyang, J. Wu, X. Jiang, Training language models to follow instructions with human feedback, in: Advances in Neural Information Processing Systems (NeurIPS), volume 35, 2022, pp. 27730–27744.
- [41] R. Jiang, K. Chen, X. Bai, Z. He, J. Li, M. Yang, T. Zhao, L. Nie, M. Zhang, A survey on human preference learning for large language models, arXiv preprint arXiv:2406.11191 (2024).
- [42] C. Comito, M. Guarascio, A. Liguori, F. S. Pisani, DALEK: combining deep active learning and explanation methods for fake news detection on COVID-19, Neural Computing and Applications 38 (2026) 19.
- [43] F. Piccoli, C. Rota, R. Kumar, G. Ciocca, PyTAGIT: a scalable and interactive human-in-the-loop tool for fast image annotation, Neural Computing and Applications 38 (2026) 226.
- [44] L. Pensel, S. Kramer, Human guided learning of transparent regression models, Knowledge-Based Systems 350 (2026) 116527.
- [45] M. M. Madduri, M. Yamagami, S. J. Li, S. Burckhardt, S. A. Burden, A. L. Orsborn, Computational framework to predict and shape human-machine interactions in closed-loop, co-adaptive neural interfaces, Nature Machine Intelligence 8 (2026) 372–387.
- [46] H. Xiao, K. Rasul, R. Vollgraf, Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms, arXiv preprint arXiv:1708.07747 (2017).
- [47] J. Yang, R. Shi, D. Wei, Z. Liu, L. Zhao, B. Ke, H. Pfister, B. Ni, MedMNIST v2 — a large-scale lightweight benchmark for 2D and 3D biomedical image classification, Scientific Data 10 (2023) 41.
- [48] A. Acevedo, A. Merino, S. Alférez, Á. Molina, L. Boldu, J. Rodellar, A dataset of microscopic peripheral blood cell images for development of automatic recognition systems, Data in Brief 30 (2020) 105474.
- [49] A. Krizhevsky, Learning multiple layers of features from tiny images, Technical Report, University of Toronto, 2009.
- [50] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016, pp. 770–778.